



Yu-Hao (Henry) Lee

Senior Software Engineer

Frontend · AI · Full-Stack Web Development

Senior Software Engineer specializing in Next.js 16 and building generative AI-powered products, with the ability to independently ship complete products.

I established a documentation-driven workflow with Claude Code that enables the delivery of one to two production-ready features per day.

I independently built AI Chat, AI Dictionary, AI Analyzer, and other products, demonstrating end-to-end product development capabilities.

I continue to explore AI-native development practices and advise teams on adopting modern engineering tools and workflows.

[Portfolio](#) | [LinkedIn](#) | [GitHub](#) | [Medium](#)

Work Experience

Senior Software Engineer

Independent / Freelancer • April 2024 - Present



Core Technologies:

- **AI Integration & Engineering:** Vercel AI SDK, Google Gemini API, Prompt Engineering, Streaming UI, Multi-Model Architecture
- **Frontend Architecture:** Next.js 16 (App Router), React 19, TypeScript, Turborepo (Monorepo)
- **UI Engineering:** Tailwind CSS, Framer Motion, shadcn/ui, DaisyUI
- **Development Workflow:** Documentation-Driven Development, Claude Code CLI

Highlights:

- **Personal Technology Platform (HenryLeeLab.com):** Built a Next.js 16 and Turborepo monorepo platform integrating AI Chat, AI Dictionary, AI Analyzer, Time Tracker, and other modules.
- **AI-Native Chat Interface (AI Chat):** Implemented multi-model switching (Gemini/GPT/Mistral), streaming responses, and context awareness to solve the UX pain points of LLM interaction latency.
- **AI-Assisted Development Workflow:** Established a documentation-driven development standard with Claude Code, validated in practice to sustain a delivery rate of 1–2 complete features per day.
- **Technical Consulting:** Led short-term engagements for front-end/back-end API contract alignment (OpenAPI) and cross-functional communication, using AI to resolve complex technical bottlenecks alongside senior backend and design teams.

Frontend Engineer

EUCARE Co., Ltd. • March 2022 - April 2024



Tech Stack: Next.js, TypeScript, styled-components, MUI, Nx (Monorepo), React Query, LINE LIFF, etc.

Tools: GitHub, ClickUp, Figma, Twilio

Highlights:

- **Cost and Workflow Optimization:** Introduced a new development approach — including adopting AI-assisted development in early 2023 — that reduced development costs by 50% and improved team development efficiency by 60%.
- **Video Consultation Integration:** Independently researched and overcame complex Twilio SDK integration challenges, designing a simpler, more stable connection flow to deliver video consultations and screen sharing.
- **LINE Official Account Services:** Built appointment booking, real-time messaging, and health information lookup using LINE LIFF, expanding the company's user service channels and improving engagement.
- **Telemedicine Platform Development:** Independently led the architecture and implementation of the full web interface (Doctor, Patient, Admin, and LINE Mini App) under a monorepo strategy, improving cross-team consistency and code reuse.
- **Technology Modernization:** Led the React version upgrade for the company's core product line and refactored 99% of the existing codebase, resolving legacy technical debt and improving stability.
- **Data Flow and Type Safety:** Redesigned frontend-backend data exchange using DTOs and adopted comprehensive TypeScript type constraints, improving data sync reliability and long-term maintainability.

Frontend Engineer

WishMobile

WishMobile, Inc. • September 2021 - March 2022

Tech Stack: React, React Hook, Redux Toolkit, Redux Saga, styled-components, Tailwind, etc.

Tools: GitHub, Jira, ClickUp, Figma, Zeplin

Highlights:

- **Modular Architecture:** Partnered with the app team to design a modular architecture, building reusable packages and libraries for rapid project assembly.
- **Engineering Standards:** Established coding standards and introduced ESLint, Prettier, and Husky, improving code review quality and consistency.
- **Documentation and Cross-Team Alignment:** Documented repository setup, folder structure, and conventions to reduce communication overhead across teams.
- **UI/UX/DX Optimization:** Improved the existing product's user experience with intuitive form patterns (linked fields, quick selection), boosting development efficiency and usability.
- **Full-Stack Collaboration:** Supported backend workflow optimization and quickly implemented design requirements from a full-stack perspective, resolving technical bottlenecks.

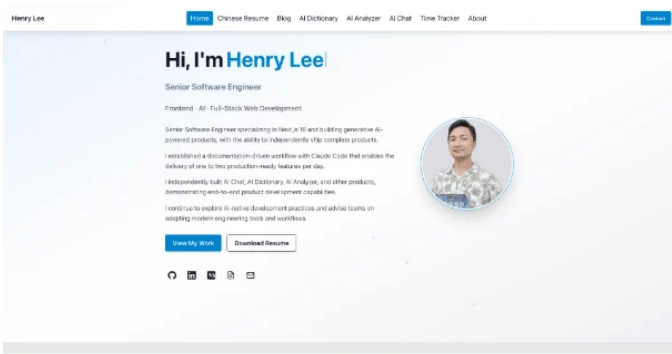
Full-Stack Engineer



Arisan Intelligence Co., Ltd. • June 2020 - May 2021

- Independently migrated the company website (www.arisan.io) from a jQuery architecture to React.
- Contributed to front-end and back-end development for the company's monitoring integration system.
- Developed a more efficient SCSS authoring approach, improving layout development efficiency by over 50%.
- Built several reusable components to streamline development.
- Collaborated to improve multi-channel video loading time from over 5 seconds to under 1 second.

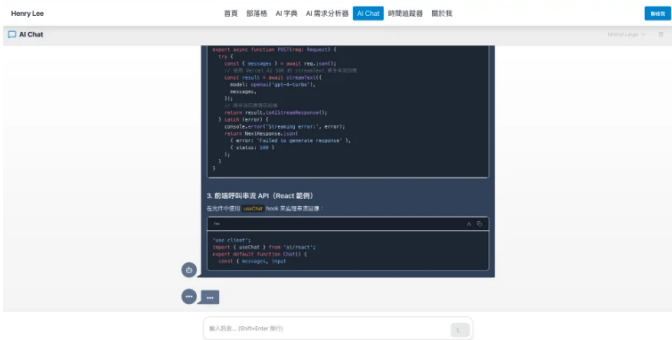
Projects



Henry Lee Personal Website — Full-Stack Personal Brand Platform

A personal website combining a resume, project portfolio, AI tools, and content automation — the core platform for showcasing my job search and full-stack development and product design capabilities.

- **Tech Architecture:** Built with Next.js 16, TypeScript, and Tailwind CSS, supporting the App Router.
- **AI Tool Integration:** Integrates three self-built AI tools — AI Chat, AI Dictionary, and AI Analyzer — connected to multiple LLMs including Google Gemini, Groq, and Mistral.
- **Content Automation:** Automatically syncs, parses, and carousels Medium articles, reducing manual maintenance.
- **Modular Presentation:** Showcases a skills matrix, project cards, and a work experience timeline through modular components.
- **Tech Stack:** Next.js, TypeScript, Tailwind CSS, React Query, DaisyUI, Shadcn UI, Google Gemini API
- Site: [Live Site](#), [Implementation Article](#)



AI Chat — Multi-Model Conversational Interface

A frontend application designed for LLM conversation experiences, solving the two major UX pain points of “single-model limitation” and “perceived response latency” to deliver a smooth AI chat experience.

- **Multi-Model Switching:** Integrates leading LLMs such as Gemini, Groq, and Mistral, letting users switch in real



AI Dictionary — AI-Powered Chinese Dictionary

An AI-powered vocabulary learning tool combining AI and etymology to help users deeply understand the meaning and origin of Chinese words.

- **Definition Generation:** AI generates easy-to-understand definitions with part-of-speech tags, ideal for language learning and education.
- **Etymology Analysis:** Uses AI to break down each character's origin, providing Zhuyin, Pinyin, and etymological explanations for more context.
- **Loanword Tracing:** Automatically identifies loanwords and traces their origin and full path into Chinese.
- **Learning Management:** Supports a “Mark as Mastered” feature to track learning progress.
- **Tech Stack:** Next.js 16, Google Gemini API, React Query, TypeScript
- Site: [Live Site](#), [GitHub](#)



AI Analyzer — AI Requirements Analyzer

Built after noticing that many people struggle to communicate effectively with AI — this “AI for AI” tool uses guided follow-up questions to help users clarify what they actually need.

- **Pain Point:** Many people don't know where to start when prompting AI, leading to answers that miss the mark.

time without being limited by a single model's quota or capability.

- **Streaming UI:** Implements a streaming interface so AI responses render word-by-word, eliminating the anxious blank wait typical of traditional AI chat.
- **Markdown Rendering:** Fully supports syntax-highlighted code blocks, lists, headings, and other formatting — ideal for technical discussion and code generation.
- **Tech Stack:** Built with Next.js 16, Vercel AI SDK, React 19, and TypeScript, deployed via Vercel.
- Site: [Live Site](#), [GitHub](#)

- **Requirement Breakdown:** AI automatically decomposes vague ideas into core elements — goals, users, features, and technology.
- **Guided Follow-Up Questions:** Generates clarifying questions for each element to help users refine their requirements.
- **Prompt Learning:** Uses structured templates to teach users how to write more precise prompts for AI.
- **Tech Stack:** Next.js 16, Google Gemini API, React Query, TypeScript
- Site: [Live Site](#), [GitHub](#)

SKILLS

AI Application Development

- Vercel AI SDK (Multi-Model Integration)
- Google Gemini API
- Prompt Engineering
- Streaming UI

Modern Frontend Architecture

- Next.js 16
- React 19
- TypeScript
- TanStack Query
- Zustand

UI and Development Tools

- Tailwind CSS / DaisyUI
- Turborepo (Monorepo)
- Claude Code-Assisted Development

Backend Experience

- Node.js / Express
- MongoDB
- Supabase

LANGUAGES

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- Chinese (Mandarin) – Native
 - English – Intermediate

EDUCATION

[Mentor Program, Cohort 3](#)

Full-Stack Engineering Training • Apr 2019 - Dec 2019

Minghsin University of Science and Technology – Bachelor's Degree

Chemical and Materials Engineering • 2009 - 2011